

Response: Here are the step-by-step solutions for the two Class Tests provided in the document.

Class Test-1 Solutions

1. Definition of Accounting and its Users

- **Accounting** is the systematic process of identifying, measuring, recording, classifying, summarizing, analyzing, interpreting, and communicating the financial information of an organization to interested users.
- **Five important users of accounting information:**
 - **Investors:** To determine the feasibility and profitability of investing in the business.
 - **Creditors/Lenders:** To assess the company's ability to pay off debts and interest.
 - **Management:** To make informed internal strategic and operational decisions.
 - **Government/Tax Authorities:** To ensure compliance with regulations and determine tax liabilities.
 - **Employees/Labor Unions:** To assess the company's stability and negotiate wages or benefits.

2. Journal Entries and Ledger for Abby Roland (July 2023)

i) **Journal Entries** Below are the general journal entries for the transactions that occurred in July 2023.

Date (July)	Account Titles and Explanation	Debit (TK)	Credit (TK)
1	Cash		
	Abby Roland, Capital	15,000	15,000
	<i>(Invested cash into the business)</i>		
2	Equipment (Van)	16,000	
	Cash		2,000
	Notes Payable		14,000

Date (July)	Account Titles and Explanation	Debit (TK)	Credit (TK)
	<i>(Purchased van, paid partial cash and signed note)</i>		
3	Rent Expense Cash <i>(Paid office rent for the month)</i>	1,500	1,500
4	Accounts Receivable Service Revenue <i>(Performed services on account)</i>	10,400	10,400
5	Abby Roland, Drawings Cash <i>(Withdrew cash for personal use)</i>	1,000	1,000
6	Supplies Accounts Payable <i>(Purchased supplies on account)</i>	850	850
7	Cash Accounts Receivable	2,000	2,000

Date (July)	Account Titles and Explanation	Debit (TK)	Credit (TK)
	<i>(Received cash for services provided on July 4)</i>		
8	Gasoline Expense Accounts Payable <i>(Purchased gasoline on account)</i>	1,200	1,200
10	Notes Payable Cash <i>(Made payment on note payable)</i>	2,500	2,500
12	Utilities Expense Cash <i>(Paid for utilities)</i>	1,250	1,250
14	Salaries Expense Cash <i>(Paid employees' salaries)</i>	2,000	2,000

ii) Ledger of Cash Account This is a standard running-balance ledger for the Cash account based on the journal entries above.

Date (July)	Explanation	Debit (TK)	Credit (TK)	Balance (TK)
1	Initial Investment	15,000		15,000

Date (July)	Explanation	Debit (TK)	Credit (TK)	Balance (TK)
2	Down payment for Van		2,000	13,000
3	Paid Office Rent		1,500	11,500
5	Owner's Drawings		1,000	10,500
7	Collection from Account	2,000		12,500
10	Payment on Note Payable		2,500	10,000
12	Paid Utilities		1,250	8,750
14	Paid Salaries		2,000	6,750

Class Test-2 Solutions

1. Definitions with Examples

- **Break-even Point:** The level of sales at which a company's total revenues equal its total expenses, meaning the company experiences neither a profit nor a loss (Net Operating Income is zero).
 - *Example:* If a company has fixed costs of 10,000 TK and a contribution margin of 10 TK per unit, selling exactly 1,000 units covers all costs. At 1,000 units, the company breaks even.
- **Contribution Margin:** The amount of sales revenue that remains after variable expenses have been deducted. It is the amount available to cover fixed expenses and then provide profits for the period.
 - *Example:* If a specialized telephone sells for 75 TK and costs 60 TK in variable expenses to produce, the contribution margin is 15 TK per unit.

2. ABC Company Analysis

Based on the provided income statement for 20,000 units, the base figures are:

- Sales: 15,00,000 TK (75 TK/unit)
- Variable Expenses: 12,00,000 TK (60 TK/unit)
- Contribution Margin: 3,00,000 TK (15 TK/unit)
- Fixed Expenses: 2,50,000 TK
- Net Operating Income: 50,000 TK

i) CM Ratio and Variable Ratio

- **CM Ratio** = Contribution Margin / Sales = 3,00,000 / 15,00,000 = **20%**

- **Variable Ratio** = Variable Expenses / Sales = 12,00,000 / 15,00,000 = **80%**

ii) **Break-even Point using Equation Method** Let Q equal the number of units sold to break even.

$$\text{Sales} = \text{Variable Expenses} + \text{Fixed Expenses} + \text{Profit}$$

$$75Q = 60Q + 250,000 + 0$$

$$15Q = 250,000$$

$$Q = 16,666.67 \text{ units}$$

- **Break-even in Units: 16,667 units** (rounding up to the nearest whole unit).
- **Break-even in Sales (Taka):** $16,666.67 \times 75 = \mathbf{1,250,000 \text{ TK}}$

iii) **Units to Earn Profit of TK 90,000 (Contribution Method)**

$$\text{Target Units} = \frac{\text{Fixed Expenses} + \text{Target Profit}}{\text{Contribution Margin Per Unit}}$$

$$\text{Target Units} = \frac{250,000 + 90,000}{15}$$

$$\text{Target Units} = \frac{340,000}{15} = \mathbf{22,667 \text{ units}}$$

iv) **Margin of Safety in Taka and Percentage**

- **Margin of Safety (TK)** = Actual Sales - Break-even Sales
 - $15,00,000 - 12,50,000 = \mathbf{2,50,000 \text{ TK}}$
- **Margin of Safety (%)** = Margin of Safety (TK) / Actual Sales
 - $(2,50,000 / 15,00,000) = \mathbf{16.67\%}$

v) **Expected Increase in Net Operating Income** First, determine the Degree of Operating Leverage (DOL):

$$\text{DOL} = \frac{\text{Contribution Margin}}{\text{Net Operating Income}} = \frac{300,000}{50,000} = 6$$

If sales increase by 8%, the expected percentage increase in net operating income is:

- **Expected Increase** = Percentage Increase in Sales $\times$ DOL = $8\% \times 6 = \mathbf{48\%}$

vii) **Verification via New Income Statement (8% Sales Increase)** If sales increase by 8%, the new sales volume is $20,000 \times 1.08 = 21,600 \text{ units}$.

Item	Calculation	Total (TK)
Sales	21,600 units × 75 TK	16,20,000
Variable Expenses	21,600 units × 60 TK	12,96,000
Contribution Margin	21,600 units × 15 TK	3,24,000
Fixed Expenses	Remains unchanged	2,50,000
New Net Operating Income		74,000

- **Increase in Net Operating Income** = 74,000 - 50,000 (Original NOI) = 24,000 TK
 - **Percentage Increase** = $(24,000 / 50,000) \times 100 = 48\%$ *(This successfully verifies the answer obtained in step v).*
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